

Machine Learning Final Report

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1. INTRODUCTION

The sun is on an 11-year magnetic cycle that varies in intensity throughout the cycle. At the beginning and ends of the cycle there are less events and an overall calmer solar climate. Whereas the peak of the cycle can cause intense solar flares and even damaging coronal mass ejections. For the current solar cycle, the cycle peak is from around 2024-2026, which explains an influx of solar activity experienced in the later half of 2025. The magnetic field lines of the sun are a lot messier than the field lines on Earth since the plasma contributing to the magnetic field is dynamic. Eventually the magnetic field gets so tangled that the magnetic poles flip and the cycle starts again [NOAA \(2020\)](#). So a full cycle would last for roughly 22 years, considering 11 years for each flip in the magnetic field.

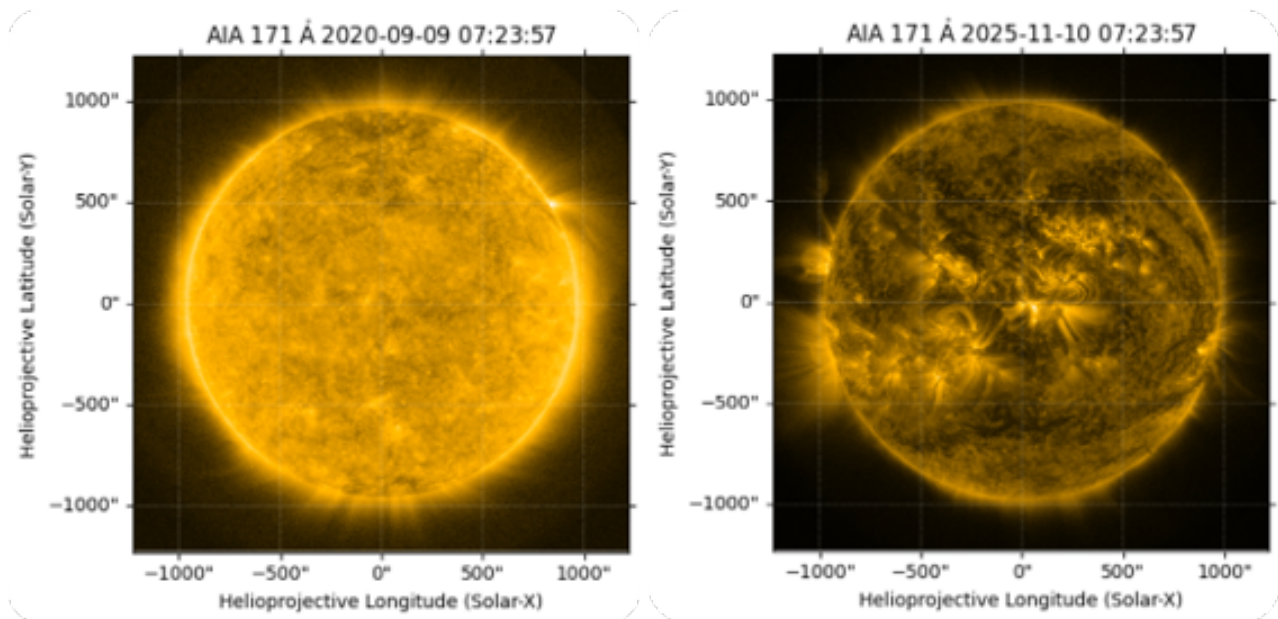


Figure 1. Map of the sun in 2020 (solar minimum) vs. 2025 (solar maximum)

2. PURPOSE

The purpose of this experiment is to detect where the sun is in its 11-year cycle given the quantity of certain events occurring on the sun in the given time. This will be done using a machine learning model and a variety of different features. Having multiple features is important because while they all peak at around the solar maximum, there is some variability to each individual feature and that variation can be used to pinpoint when exactly the sun is in its cycle. For example, eruptive events peak towards the end of the cycle, slightly after the solar event maximum. Whereas oscillations have three separate peaks around the middle of the cycle. This model can be used as an alternative way to measure where the sun is in its cycle, and perhaps with modification can be applied to other stars once technology is advanced enough to measure the number of features coming from the targeted star.

3. METHODS

3.1. Data Methods

There are python libraries designed to help with astronomical data and machine learning strategies. The main library I used for this project was Sunpy to query data, Sklearn to use a machine learning model and predict data, Matplotlib to chart data and images, and Numpy/Pandas to organize and format data. Sunpy was a very helpful library for this project, it contained methods to query data and images from the sun from many different solar observatories.

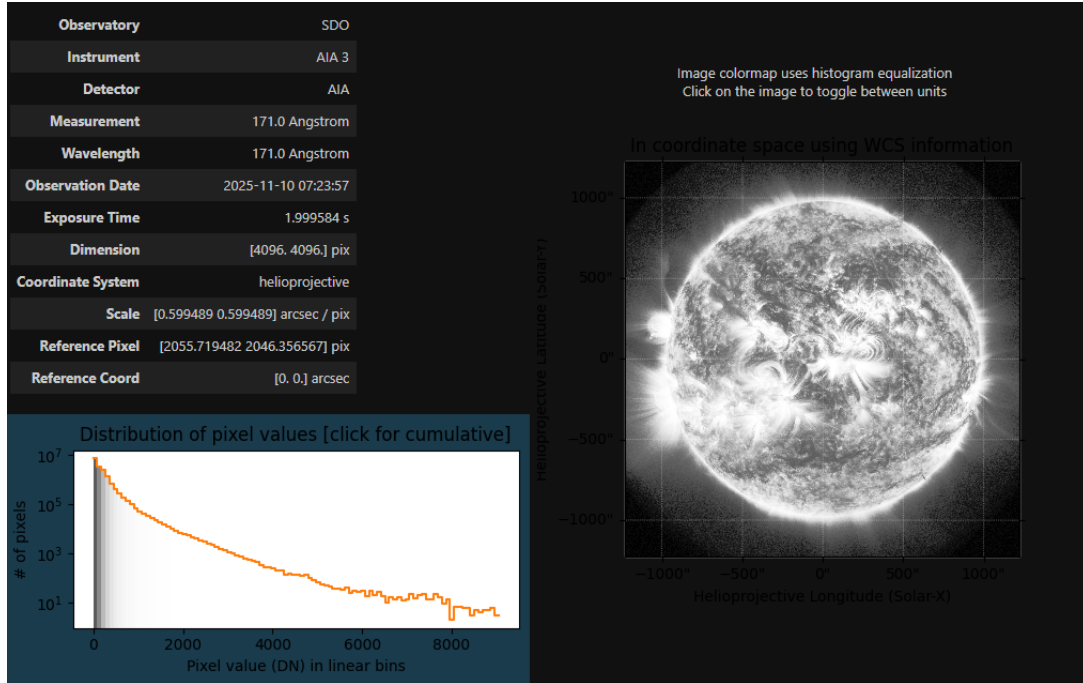


Figure 2. Example Sunpy map with information about the sun from the queried data

The website I used to gather data was the Heliophysics Event Knowledgebase (HEK). Using Sunpy and HEK, I queried all event data from the solar cycle from December 2008 to December 2019. I was only able to take data from one full cycle because there was not enough available data from the 1997-2008 cycle and the 2019-2030 cycle hasn't completed yet. I decided to query data once every month, for a total of 132 datapoints (11 years x 12 months). The test data was queried in the same way, but took data roughly once every year, cutting datapoints at the beginning and end so there was no overlap in test and training data which would produce the same exact values.

AR - Active Regions ER - Eruption EF - Emerging Flux SG - Sigmoid FL - Flare FE - Filament Eruption FA - Filament Activation CME - Coronal Mass Ejection CH - Coronal Hole CD - Coronal Dimming CW - Coronal Wave	FI - Filament PG - Plage SP - Sunspot SS - Sunspot Group OS - Oscillation TO - Tornado/Prominence CR - Cosmic Rays CE - Coronal Jet EE - Eruptive Event NR - Nothing Reported XX - Miscellaneous
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Figure 3. All features provided by HEK

There are many events that are available from HEK. Some of the features were omitted from the feature count because they were either too rare to provide meaningful data, were breaking the program, or just used as a placeholder. The 15 features chosen were: [AR, ER, SG, FL, FE, FA, CH, CW, PG, SP, SS, OS, CR, CE, EE].

Since some features are rarer than others, they each have a different range of time to use to query data. The data is collected by observing the quantity of events that occurred within the specific time frame for each feature, centered around the single time imputed to the function. For example, a flare is common, so it has a time range of six hours. So if the method is inputted 11-29-2016 00:00:00, it will check from 11-28-2016 18:00:00 to 11-29-2016 06:00:00 and return the amount of flares that occurred in that time frame. Some features were providing abnormal results, so they were denoted as having a range of one hour but were omitted from the actual feature list. The time range was setup to get around 5-50 features per query. This will still provide meaningful data but keep the time obtaining data relatively small. Since the features do not require feature scaling, it is not pivotal for them to be scaled to other features, only themselves.

AR - 1 Hour ER - 1 Day EF - 1 Month SG - 1 Day FL - 6 Hours FE - 15 Days FA - 2 Months CME - 1 Hour CH - 1 Hour CD - 1 Month CW - 3 Months	FI - 6 Months PG - 1 Year SP - 2 Months SS - 6 Hours OS - 2 Months TO - 1 Day CR - 2 Months CE - 1 Day EE - 6 Months NR - 1 Hour XX - 1 Hour
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Figure 4. Delta time for all features

3.2. Machine Learning Methods

The machine model used will be a gradient boosting regressor provided by Sklearn. A booster is ideal for this project because there the feature to data-point ratio is relatively large (15/132). If a bagging method was used there would be data-points that would not be used, and with a small number of data-points, I cannot afford to lose any of them. Whereas a booster will iteratively improve the model with the small number of data-points available.

The hyperparameters modified for this model was the 'learning_rate', 'n_estimators', and 'max_depth'. After some tests, the optimal values for these hyperparameters are 0.05 for the learning rate, 70 for n estimators, and 2 for max depth.

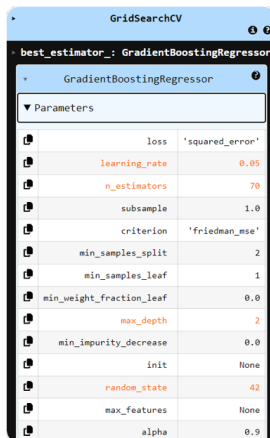


Figure 5. GridSearch for my optimal model

4. DATA

The data obtained was as expected, most features peaked in the middle of the cycle and contained less features at the beginning and ends of the cycle. Although there were some features that peaked multiple times, and others that peaked towards the end of the cycle.

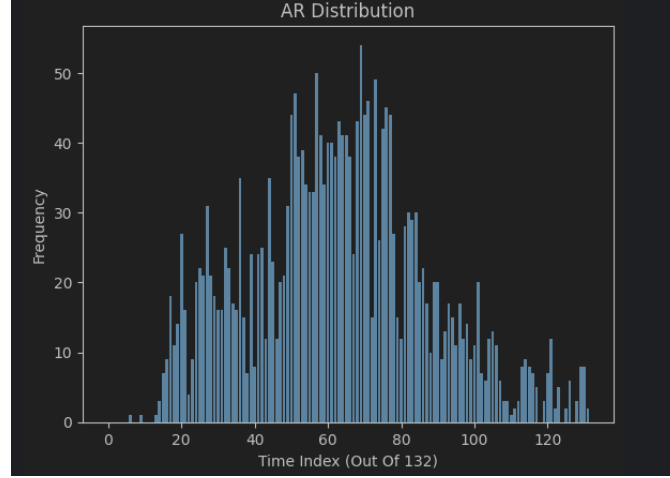


Figure 6. Distribution of the quantity of active regions on the sun throughout the 2008-2019 solar cycle

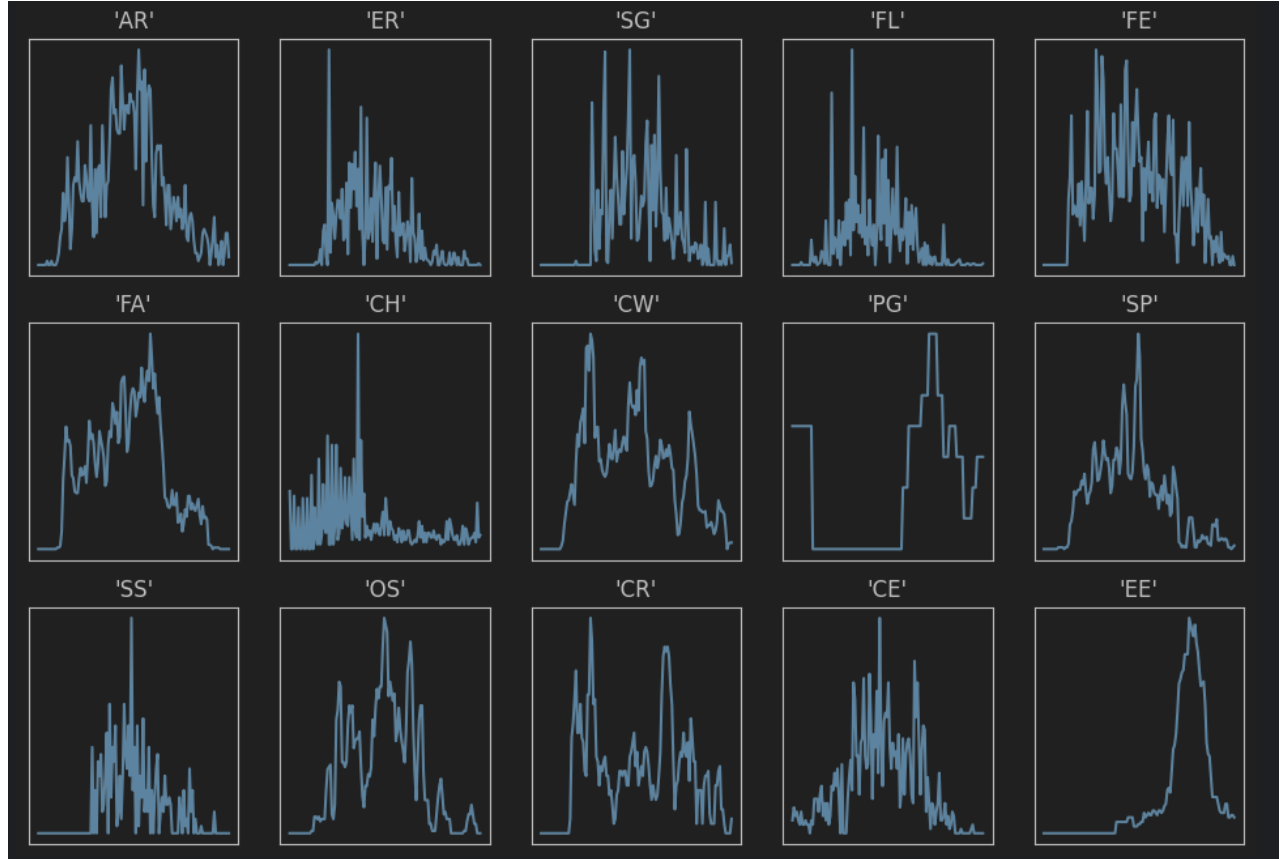


Figure 7. Distribution of the quantity of all features on the sun throughout the 2008-2019 solar cycle

5. RESULTS

5.1. *Scoring Results*

The results were surprisingly high at a score of around 99.3%. And an average difference from prediction to result of 66.2 days, which is relatively small when considering a span of 11 years.

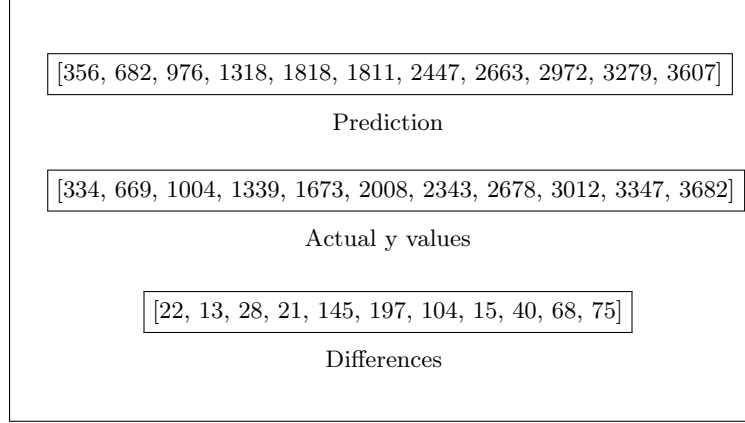


Figure 8. Machine learning model prediction results

Unfortunately the results were not as high with data from 2019-2025. The model was predicting that it was in the end of the last cycle, instead of the beginning of the next cycle. To the point the point where the score I got for the current data was at -11.

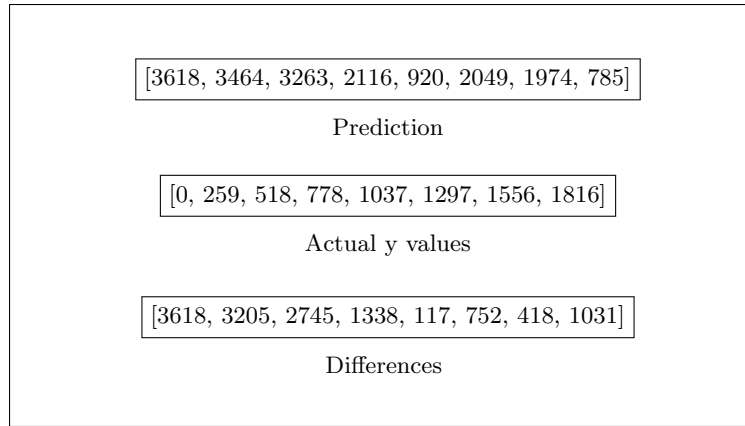


Figure 9. Machine learning model prediction results for 2019-2025 data

5.2. Feature Importance

The feature importance results consisted of [EE] holding the most importance, [FA, OS, SP, SG, CE, CR, AR] holding some importance, and [FE, PG, CW, ER, FL, SS, CH] holding no importance.

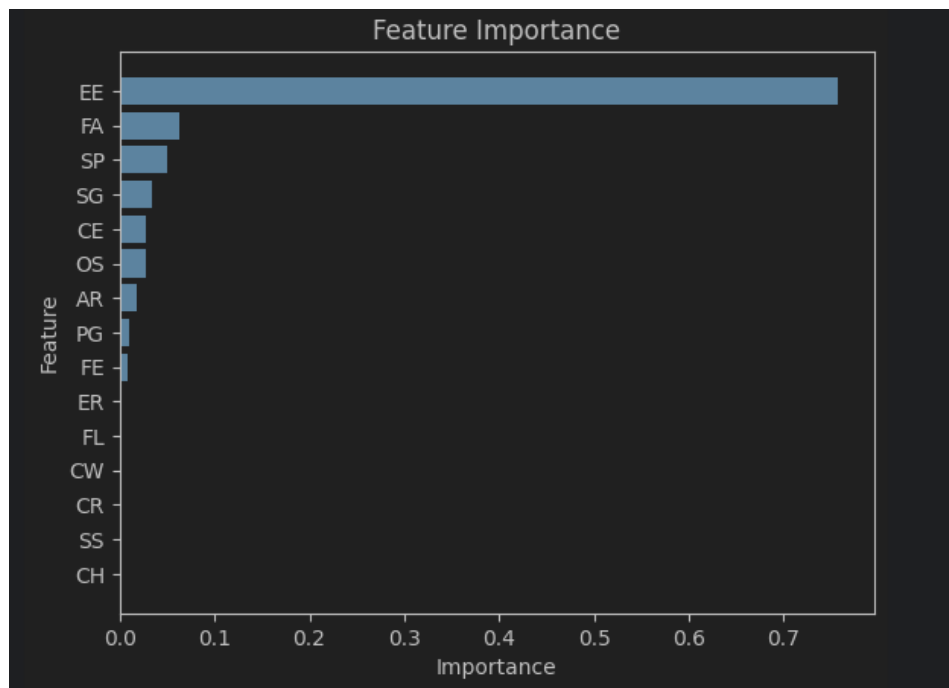


Figure 10. Feature importance bar graph

6. CONCLUSION

6.1. *Discussion*

6.1.1. *Scoring Results*

The results were very positive for the 2008-2019 data. An accuracy score of over 99% is very good. However, I believe that this high accuracy was due more in part to the fact that it was memorizing the specific 2008-2019 cycle rather than using actual data to predict things. This is reinforced by the fact that there was a low accuracy for the current test data.

6.1.2. *Feature Importance*

The feature importance was personally surprising. Flares had no impact in the prediction. Looking at the graphs and feature importance, it seems like any value that has an abnormal peak is a good predictor. For example, eruptive events hold the most influence by far, and they peak later in the cycle than most other features. This is a very interesting trait and one I may use for future research. Perhaps using only the important features I can get better results and use more models that work well with a low feature to data-point ratio.

I also found it surprising that the delta time did not affect the feature importance very much. I believed that features with a greater delta time would have more significance in predicting the cycle because they would more or less remain constant in shorter time intervals. The same Filament that happened once will appear in the data for 12 months. But while that held some truth, there were a lot of features with significance that have a delta time of less than 1 day.

6.2. *Error Report*

This is a very static model that cannot be used to detect the progress in any cycle of the sun. Since the 2019-2030 solar cycle is still ongoing, and the 1997-2008 cycle doesn't have as much data provided due to older technology, I was only able to build the model around a single solar cycle. I feel that this may have allowed and encouraged the model to overfit and I believe that is why I have such a high accuracy for 2008-2019 data, but such a low accuracy for 2019-2025 data. However, I think part of the low accuracy for the current test data is because it was predicting a time at the end of the last cycle, which may have only been months away from the correct time, yet the model marked it as very far from the correct time because it was at the other end of the cycle. Another explanation could be that there is variability in the cycles when the poles are flipped. Maybe a 22-year cycle is needed to properly model such a system. Regardless, in the future I would like to make a model that can predict the progress in any given solar cycle, not just the one from 2008-2019.

There was also error in the querying of data. Some telescopes or events would throw errors at random points in time and break the program altogether. Some issues were caused by outdated telescopes, and others because it was reporting an altitude of $> 90^\circ$. Even though there was no need for the altitude in this project, there was no way to partially query data. This was not very common, and only occurred for a few features for a few data-points, so I returned a value of 0 for the current feature count. Although unlikely, it is possible that this had an impact on the results.

6.3. *Conclusion*

In conclusion, this project is very good at predicting where in the 2008-2019 cycle the sun is in given the amount of features currently active on the sun. However, it is sub-par at predicting a general abstract cycle. In the future I would like to work on making a more general model that can predict dynamic solar cycles, but for that I need more data from the sun. Additionally I would like to make the data cyclic and only test significant features to get better results with more data-points and less features. And maybe one day modify this model to predict the cycles of other stars. But overall I am satisfied with the results and proud of what I have accomplished.

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